Error Analysis — the "menstrual" interpretation gap

Question raised by the baseline report: the base model interprets menstrual concerns at only ~0.39 while fertility is ~0.90 (§4.3). Is this a model blind spot, or a labeling artifact? **Answer: predominantly a category-boundary artifact — the "errors" are largely clinically-defensible reads of inherently overlapping cases.**

All figures from `M2ml_{en,fr,ar}.jsonl` scored by `scripts/evaluate.py`.

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1. What menstrual-gold items get predicted as

Confusion for the 180 menstrual-gold items per language (base model M2, parse_ok):

Predicted	EN	FR	AR
menstrual (correct)	0.39	0.39	0.40
fertility	0.38	0.36	0.36
PCOS	0.14	0.20	0.20
other	0.08	0.04	0.04

The misreads are not random: **87.2%** of the 109 wrong EN menstrual items are mapped to an **adjacent category (fertility or PCOS)**; only **12.8%** go to "other". The model isn't failing to understand — it is disambiguating an overlapping case toward a different, usually reasonable, category.

2. Why: the menstrual seeds are conception- and PCOS-themed

14 of 30 menstrual seeds (47%) explicitly mention conception/pregnancy. These are labeled `menstrual` (presenting symptom = irregular periods) but their *chief complaint / patient intent* is fertility. Actual inputs the model "got wrong":

Seed	Gold	Model	Input (verbatim)
menst-002	menstrual	fertility	"Do irregular periods influence the ability to get pregnant?"
menst-008	menstrual	fertility	"Due to my irregular periods I am worried about getting pregnant … Is it possible for me to conceive even with irregular periods?"
menst-018	menstrual	fertility	"married for 4yrs, but i havnt conceived, my menstrual cycle is sometimes painful … desperate need a child"
menst-019	menstrual	fertility	"missed my periods by 3 days and the test on the strip came weakly positive … could I be pregnant?"
menst-021	menstrual	fertility	"35 year old married woman … trying for child since 1 yr … i missed periods …"

Seed	Gold	Model	Input (verbatim)
menst-023	menstrual	fertility	"want to be pregnant because of my irregular periods my dr suggest me to take siphene …"
menst-030	menstrual	**PCOS**	"irregular periods. After an ultrasound … I have multiple cysts over the periphery of ovaries. I want to conceive …"

- The **fertility** confusions are conception-motivated period complaints — the model keys on the stated goal ("want to be pregnant", "trying to conceive") rather than the presenting symptom. Clinically this is a defensible, arguably more useful, read.
- The **PCOS** confusions (e.g. menst-030) are textbook PCOS presentations — *irregular periods + polycystic ovaries on ultrasound* — where the label `menstrual` is debatable and `PCOS` is arguably **more correct** than the gold.

12 menstrual seeds are wrong on **all 6 phrasings** in every language — i.e. it is a stable property of the *case/label*, not of surface form or language.

3. Cross-model: fine-tuning shifts the confusion, doesn't fix it

Model	Lang	menstrual interp	dominant wrong-prediction
M2 (base)	EN/FR/AR	0.39 / 0.39 / 0.40	**fertility** (69/65/65)
M3ml-v1	FR/AR	0.37 / 0.38	**PCOS** (83/80)
M3ml-v2	EN/FR/AR	0.37 / 0.33 / 0.38	**PCOS** (84/82/74)

Fine-tuning **moves the menstrual confusion from fertility (base) toward PCOS**, because the adaptation corpus reshapes the model's category priors — but the underlying overlap is not resolved by any model. This is itself a finding: the low menstrual score is a property of the **taxonomy/labels interacting with real patient language**, robust across models and languages.

4. Implications for the benchmark and paper

1. **Raw category accuracy understates true interpretation quality.** ~87% of menstrual "errors" are clinically-adjacent, and a non-trivial share (the conception- and PCOS-themed seeds) are reasonable or arguably correct.
2. **Real patient messages do not respect clean category boundaries.** Half the menstrual seeds are conception-motivated; irregular-periods-with-ovarian-cysts is simultaneously "menstrual" and "PCOS". This is a genuine language-understanding phenomenon, not just noise.
3. **Recommended actions for the paper (pick one or combine):**
   - Report a **relaxed / clinically-acceptable interpretation** metric that credits adjacent-category reads (menstrual↔fertility when conception is the stated intent; menstrual↔PCOS when polycystic ovaries are described), alongside strict accuracy.
   - **Refine the gold labels** for the ~14 conception-motivated menstrual seeds (multi-label, or relabel by chief complaint), then re-score.
   - Keep strict accuracy but **explicitly frame the menstrual gap as category-boundary ambiguity** with this confusion analysis as evidence — turning a weak-looking number into a substantive finding.
4. **Do not** claim the base model "cannot understand menstrual concerns" — the evidence contradicts it. The correct claim is that menstrual presentations in real patient language frequently carry fertility or PCOS intent that the model (reasonably) surfaces.

5. Relaxed (clinically-acceptable) interpretation metric

We implemented the recommended relaxed metric (`scripts/relaxed_interp.py`, unit tests in `tests/test_relaxed_interp.py`). A prediction is credited if it lands in a per-item **acceptable set** = {gold} ∪ {clinically-adjacent categories *justified by the case content*}. Adjacency is **content-gated**, not blanket: an adjacent category counts only when the case text contains its clinical markers (conception/ pregnancy → fertility; cysts/polycystic/ovaries → PCOS; period/cycle/bleeding → menstrual). Markers are detected on the **English source per seed** (case content is language-independent) and applied uniformly across EN/FR/AR.

Three metrics, bracketing interpretation from exact-match to upper bound: - **strict** — `predicted == gold` (exact match). - **relaxed (gated)** — credits adjacent reads *justified by the case content* (the defensible metric). - **loose (upper bound)** — credits *any* clinical-category prediction regardless of content (only off-taxonomy `other` fails); reported to bracket the estimate.

Model	Lang	strict	**relaxed (gated)**	loose (upper)	menstrual strict→relaxed
M2 (base)	EN	0.617	**0.878**	0.930	0.394 → 0.811
M2 (base)	FR	0.614	**0.881**	0.952	0.394 → 0.800
M2 (base)	AR	0.617	**0.883**	0.952	0.400 → 0.806
M3ml-v1	FR	0.605	0.803	1.000	0.374 → 0.592
M3ml-v1	AR	0.618	0.805	0.998	0.378 → 0.600
M3ml-v2	EN	0.644	0.806	0.996	0.367 → 0.583
M3ml-v2	FR	0.617	0.810	0.994	0.333 → 0.589
M3ml-v2	AR	0.646	0.829	0.996	0.380 → 0.631

Reading the loose column. It is near-ceiling for the fine-tunes (~0.99–1.00) because they *never abstain to* `other` — they always commit to one of the three clinical categories — whereas the base model outputs `other` for ~5–7% of items. So the loose metric mostly measures "does the model ever say other," making it a weak discriminator; the **gated relaxed metric is the one to report as the headline interpretation quality**, with strict and loose as the lower/upper brackets.

Two findings:

1. **True interpretation quality is far higher than strict accuracy implies.** The base model's clinically-acceptable interpretation is ~**0.88** (not 0.62), and menstrual rises from ~0.39 to ~0.81 — confirming most "menstrual errors" are justified adjacent reads.
2. **The strict-accuracy ranking flips under clinical gating.** By strict accuracy, M3ml-v2 leads (EN 0.644 > base 0.617). By the *content-gated relaxed* metric, the **base model leads** (EN 0.878 > v2 0.806). The reason is mechanistic: the base model's menstrual errors go to **fertility**, which the conception-themed seeds *justify* (credited); fine-tuning shifted those errors to **PCOS**, which those same seeds usually do *not* justify (not credited). So the fine-tune's strict-accuracy gain partly reflects a category-prior shift toward a *less-justified* adjacent label, not better clinical understanding. This is a caution against reading the strict-accuracy improvement as a genuine interpretation gain.

Reproduce:

```
python scripts/relaxed_interp.py \
  --dir Used_Datasets/Consolidated_Datasets/200_Seed_Dataset \
  --model M2ml --model M3ml --model M3ml_v2 --langs en,fr,ar
```

6. Reproduce (confusion + examples)

```
# confusion + examples were produced with scripts/evaluate.py scorers over
# Used_Datasets/Consolidated_Datasets/200_Seed_Dataset/M2ml_{en,fr,ar}.jsonl
# (gold_category vs predicted_category on gold_category == "menstrual").
```